LTM PROJECTS
ZEROTH REVIEW

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Project 1- EduTrace - Exam-Aware Personalized RAG

Problem Statement

Students use AI tools for learning, but these tools provide generic answers and do not consider:

- Student strengths and weaknesses
- Exam syllabus and important topics
- Previous year question patterns
- Personalized study requirements

Objectives

- Analyze student performance and identify weak areas.
- Provide exam-aware learning support.
- Retrieve relevant content from notes and question papers using RAG.
- Generate accurate and personalized answers.

Dataset

- Student Data: Marks, attendance, quiz scores.
- RAG Knowledge Base: Syllabus PDFs, previous-year question papers, study notes & PPTs.

Technical Stack

- Frontend: Streamlit
- Backend: Python, Flask
- RAG: LangChain, FAISS/ChromaDB
- LLM: GPT/Llama
- Database: SQLite

Expected Outcomes

- Personalized study recommendations
- Exam-focused learning support
- Multilingual question answering
- Improved learning efficiency
- Better exam performance

Project 2 - AI Agent for Health Insurance Claim Assistance

Problem Statement:

Health insurance claim processing is time-consuming due to manual verification, document checks, and claim tracking.

Objectives:

- Automate claim assistance using an AI chatbot.
- Verify eligibility and documents.
- Detect fraudulent claims.
- Provide real-time claim updates.
- Improve customer experience.

Dataset:

- Synthea-generated healthcare and insurance claim data.
- Used for claim validation, eligibility checks, and fraud detection.

Tech Stack:

- Frontend: Streamlit / React
- Backend: Python
- **AI Model:** Azure OpenAI / GPT
- Agent Framework: CrewAI / LangGraph
- Database: PostgreSQL / SQL Server
- OCR: Azure Document Intelligence

Expected Outcomes:.

- Automated claim support.
- Reduced manual effort.
- Better fraud detection.
- Real-time claim tracking.